THE SIMON85 PROJECT

SIMON85

Pocket Guide

Graphics extension for HP-85 BASIC

Option ROM 0102 (octal)
Source: SIMON85_rom.asm, v0.13a2

An independent HP-85 extension reference

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Getting Started

SIMON85 extends HP-85 BASIC with drawing, filled shapes and pixel operations. It is independent of PRINTER85. **EBTKS** Copy the built SIMON85 option-ROM image (ROM 0102 octal) to the EBTKS SD-card `/roms85` directory and enable the ROM in the existing EBTKS `CONFIG.TXT`. A reboot/power cycle loads the option ROM. The HP Printer/Plotter ROM 0360 (octal) must also be installed. **Test** Enter `SVER`. Then try:

```
GRAPH
SCALE 0,100,0,100
CIRCLE 50,50,25,25,1
```

Syntax

Numbers may be BASIC numeric expressions. Coordinates obey the current graphics scale. The final parameter `f` means **0** clear, **1** set and **2** XOR. Use the native `GRAPH`, `GCLEAR` and `SCALE` commands to prepare the display.

Commands

SVER

```
SVER
```

Reports the installed SIMON85 ROM version.

```
SVER
```

BLOCK

```
BLOCK x1,y1,x2,y2,f
```

Draws a filled rectangular block bounded by two opposite corners. The drawing mode is specified by `f`.

```
BLOCK 10,10,40,30,1
```

HIRES

```
HIRES j,k
```

Clears the graphics screen, selects the full 320 by 200 plotting scale (with Y inverted), then sets HP BASIC's PEN to `j`. The second argument `k` is accepted for compatibility but ignored.

```
HIRES 1,0
```

LINE

```
LINE x1,y1,x2,y2,f
```

Draws a line segment between the two specified endpoints using the requested pixel mode.

```
LINE 10,10,85,70,1
```

REC

```
REC x,y,width,height,f
```

Draws a rectangle from an origin and its width and height.

```
REC 20,20,35,25,1
```

ARC

```
ARC mx,my,ws,we,ww,rx,ry,f
```

Draws an elliptical arc centered at `mx,my`; angle parameters describe the arc and radii control its dimensions.

```
ARC 50,50,0,90,1,30,20,1
```

CIRCLE

```
CIRCLE x,y,rx,ry,f[,n]
```

Draws an ellipse or circle with center and horizontal/vertical radii. Optional `n` controls the approximation/segment count.

```
CIRCLE 50,50,20,20,1
```

ANGL

```
ANGL x,y,w,rx,ry,f
```

Draws a radial line from center `x,y` to the perimeter position given by angle `w` and radii `rx,ry`.

```
ANGL 50,50,45,25,25,1
```

PAINT

```
PAINT x,y,f
```

Flood-fills the bounded region containing the starting point, using the requested pixel operation.

```
PAINT 30,30,1
```

PLOTP

```
PLOTP x,y,f
```

Sets, clears or XORs one scaled graphics pixel. Named `PLOTP` to avoid collision with HP BASIC's `PLOTTER` keyword.

```
PLOTP 10,10,1
```

TEST

```
TEST(x,y)
```

Returns 1 if a mapped CRT pixel is set; otherwise returns 0, including points outside the screen. CRT-

only; no plotter output.

```
PRINT TEST(10,10)
```

Quick Example

```
10 GRAPH
20 GCLEAR
30 SCALE 0,100,0,100
40 REC 10,10,70,70,1
50 CIRCLE 45,45,20,20,1
60 PAINT 45,45,1
70 PRINT TEST(45,45)
```

If a shape looks different with a nondefault SCALE, check the selected graphics scale and drawing mode before assuming a hardware fault.